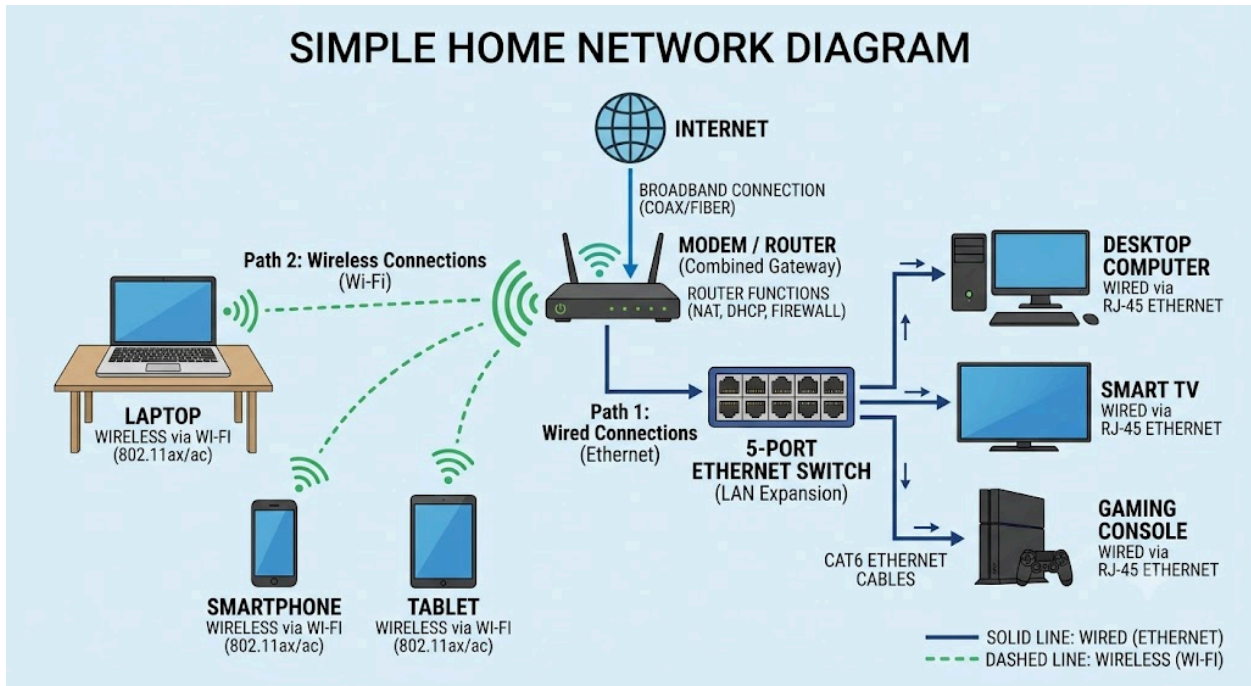


NETWORK DEVICES

Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

ANS:



Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

ANS:

Device	Primary Function	OSI Layer	Real-World Daily Use Case
Router	Connects different networks (e.g., LAN to WAN), routes IP packets, and manages IP addressing/NAT	Layer 3 (Network)	Streaming live sports or videos on YouTube —translates private home IP addresses to a public IP to fetch content from external servers.

Switch	Connects multiple devices within the <i>same</i> local network and forwards data using MAC addresses	Layer 2 (Data Link)	Multi-pc LAN gaming (e.g., <i>Valorant</i> or <i>Counter-Strike</i>)—enables instant, low-latency packet switching directly between local PCs without clogging the main router.
Hub	Legacy device that broadcasts incoming electrical signals to <i>all</i> connected ports without filtering	Layer 1 (Physical)	Monitoring legacy network traffic or connecting basic IoT sensors—broadcasting incoming data indiscriminately to every port.
Wireless Access Point (AP)	Extends a wired network by converting Ethernet signals into radio frequencies (Wi-Fi)	Layer 1 & Layer 2	Scrolling Instagram or ordering food via Zomato on a smartphone from a bedroom far from the main router without running physical cables.

Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

ANS:

Here is a step-by-step scenario explaining how your order payload travels from your phone to Zomato's cloud servers:

Scenario: Placing an Order on Zomato

You tap "**Place Order**" in the Zomato app while connected to your home network. Your phone turns that tap into a set of digital data packets containing your items, delivery address, and payment confirmation.

1. **Smartphone to Wireless Access Point (Layer 1 & Layer 2):**
 - Your phone converts the binary data into radio frequency (Wi-Fi) signals.

- The **Wireless Access Point (AP)**—often built directly into your home Wi-Fi gateway—receives these radio signals through its antennas, converts them back into electrical signals, and packages them into Ethernet frames (using MAC addresses).
- 2. **Access Point to Local Switch (Layer 2):**
 - The internal **Switch** inside your router receives the Ethernet frames on a local port.
 - It inspects the destination MAC address to forward the frames quickly across the internal wired board straight to the main router processor, keeping traffic organized and preventing data collisions.
- 3. **Switch to Router (Layer 3):**
 - The **Router** strips off the Layer 2 Ethernet headers to inspect the target **IP address** (Zomato's API gateway).
 - It performs **Network Address Translation (NAT)** to change your phone's local private IP address into your public home IP address.
 - The router then determines the optimal path across the internet and sends the packets through your modem or optical network terminal (ONT) onto your ISP's fiber/copper line.
- 4. **The Internet to Zomato's Servers (Layer 3 – Layer 7):**
 - ISP routers route your IP packets across global fiber networks until they reach Zomato's cloud servers (e.g., AWS or Google Cloud).
 - Once the packets arrive, Zomato's server reassembles the payload at the **Application Layer (Layer 7)**, confirms your order, processes payment, and notifies both the restaurant and a nearby delivery partner.

Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

ANS:

Here are two effective options to connect your gaming PC to the Wi-Fi router over a long distance:

Option 1: Powerline Adapter Kit

- **How It Works:** A powerline kit includes two adapters that plug directly into your home's electrical wall outlets, using your existing electrical wiring as a high-speed copper network cable.

1. You plug **Adapter A** into a wall socket near your Wi-Fi router and connect it to the router using a standard Ethernet cable.
 2. You plug **Adapter B** into a wall socket near your gaming PC and run an Ethernet cable straight into your PC's Ethernet port.
- **Why It Works Well for Gaming:** Unlike wireless signals that can be degraded by thick walls or distance, powerline adapters establish a direct physical link. This yields lower, more consistent ping times and minimal packet loss compared to standard Wi-Fi.

Option 2: Wi-Fi Range Extender / Wireless Access Point (with Ethernet Port)

- **How It Works:** A Wi-Fi extender or a router configured in "Client/Bridge Mode" receives wireless signals from your primary Wi-Fi router and bridges them to its built-in physical Ethernet ports.
 1. Place the extender roughly halfway between your main router and your gaming PC (where it can still catch a strong 5 GHz Wi-Fi signal).
 2. Connect your gaming PC directly to the Ethernet port on the extender using a Cat6 cable.
- **Why It Works Well:** It acts as a dedicated high-gain Wi-Fi receiver for your PC, taking advantage of larger external antennas than a standard USB Wi-Fi dongle without requiring you to open up your PC case to install a PCIe Wi-Fi card.